

Potential of Mean Force Calculation for Potassium Ion Conduction Through the KcsA Channel Using Umbrella Sampling in GROMACS

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Abstract

Ion channels are essential for regulating ion flux across biological membranes, with potassium channels like KcsA providing a canonical system for mechanistic study. In this project, I aimed to reproduce the potential of mean force (PMF) calculations from the study by Bernèche and Roux, 2001^{Simon Bernèche, 2001}, which quantified the free energy profile of K conduction through the KcsA selectivity filter using umbrella sampling. I carried out molecular dynamics simulations in GROMACS, applying a harmonic biasing potential to generate overlapping sampling windows spaced at different intervals. However, it is done through CHARMM in original paper. Initial configurations were obtained via pulling simulations, followed by short equilibration and umbrella sampling in each window. WHAM was used to reconstruct the PMF. While challenges such as non-uniform spaced intervals, multiple simulation run to fine tune pulling force led to some inaccuracies (e.g., unrealistic energy spikes, my results generally replicated key features of the original PMF profile, including the presence of energy wells corresponding to ion binding sites. This study highlights both the strengths and practical limitations of 1-D umbrella sampling.

Index Terms: Umbrella Sampling, , WHAM.

1 Introduction

Ion channels are membrane proteins that regulate the selective passage of ions across cell membranes, playing essential roles in nerve signal transmission, muscle contraction, and osmoregulation. Among these, potassium (K⁺) channels have especially been studied as a result of their high ion selectivity and rapid conduction rates. The KcsA channel (Figure 1) from *Streptomyces lividans* has served as a model system for understanding ion permeation at atomic resolution, due to its available crystal structures and relatively simple tetrameric architecture.

The conduction of ions through such channels is driven by electrochemical gradients, but is also highly regulated by the structural and energetic landscape of the selectivity filter. To map this energy landscape, potential-of-mean-force (PMF) calculations are commonly employed. PMF profiles describe the free energy associated with moving ions along the conduction pathway and reveal crucial information about the binding sites and energy barriers.

In their 2001 study, Bernèche and Roux performed umbrella sampling in molecular dynamics (MD) simulations to calculate the PMF of two K⁺ ions moving through the selectivity filter of the KcsA channel. Their results quantitatively captured the energetics of ion conduction and offered deep insight into the coupling between ion positions and hydration/dehydration events.

I tried to reproduce key features of their PMF calculation using GROMACS and 1D umbrella sampling. I simulate the motion of a single K⁺ ion along the conduction axis (Z-axis) of the selectivity filter, applying harmonic biasing potentials at discrete positions to ensure adequate sampling. Using the Weighted Histogram Analysis Method (WHAM)^{Kumar et al., 1992}, I reconstruct the PMF and compare it with the trends observed in the original study. This experiment also highlights practical aspects of umbrella sampling, including the importance of window overlap, bias strength, and sampling time.

2 Umbrella Sampling^{Torrie et al., 1977}

In molecular simulations, calculating the free energy difference between two stable states (e.g., states A and B. See Fig 2) separated by a high energy barrier. It is difficult to sample all the coordinates in conventional molecular dynamics (MD) due to insufficient sampling across the barrier. Umbrella sampling is a powerful enhanced sampling method that enables reconstruction of the potential of mean force (PMF) along a chosen reaction coordinate ξ .

To improve sampling, we apply a series of overlapping harmonic biasing potentials $\omega(\xi)$ centered at various points $\xi_0^{(i)}$ along the reaction coordinate. The biasing potential typically has the form:

$$\omega_i(\xi) = \frac{1}{2}k_i \left(\xi - \xi_0^{(i)} \right)^2 \quad (1)$$

where k_i is the force constant for window i , and $\xi_0^{(i)}$ is the restraint center for that window. This allows the system to explore regions of the reaction coordinate that are otherwise rarely sampled.

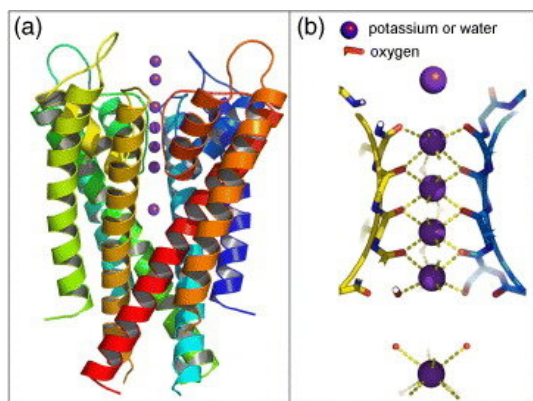


Figure 1. Schematic illustration of the KcsA potassium channel after PDB 1K4C, showing the KcsA protein complex with four transmembrane subunits (a) and selectivity with four axial trapping sites formed by the carbonyl oxygen atoms in which a potassium ion or a water molecule can be trapped (b). <http://dx.doi.org/10.1088/1367-2630/13/11/113030>

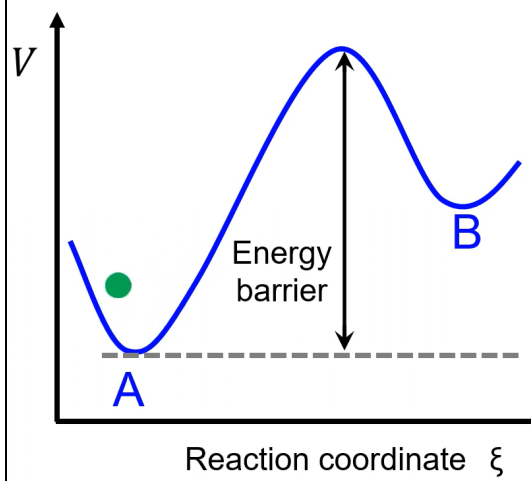


Figure 2

The biased probability distribution in window i is defined as:

$$p_i^b(\xi) = \frac{\int \delta[\xi(q) - \xi] e^{-\beta(V^u(q) + \omega_i(\xi(q)))} dq}{\int e^{-\beta(V^u(q) + \omega_i(\xi(q)))} dq} \quad (2)$$

where $V^u(q)$ is the unbiased potential energy, and $\beta = \frac{1}{k_B T}$ is the inverse thermal energy. The delta function $\delta[\xi(q) - \xi]$ ensures that only configurations at a specific ξ are counted.

This expression can be rearranged as:

$$p_i^b(\xi) = e^{-\beta \omega_i(\xi)} \cdot \frac{\int \delta[\xi(q) - \xi] e^{-\beta V^u(q)} dq}{\int e^{-\beta(V^u(q) + \omega_i(\xi(q)))} dq} \quad (3)$$

Hence, the unbiased distribution $P^u(\xi)$ is related to the biased one by:

$$P^u(\xi) = p_i^b(\xi) \cdot e^{\beta \omega_i(\xi)} \cdot C \quad (4)$$

where C is a normalization constant. The potential of mean force (PMF), $G^u(\xi)$, is then obtained from the unbiased probability as:

$$G^u(\xi) = -k_B T \ln P^u(\xi) + F \quad (5)$$

Substituting the relation between $P^u(\xi)$ and $p_i^b(\xi)$ gives:

$$G^u(\xi) = -k_B T \ln p_i^b(\xi) - \omega_i(\xi) + F \quad (6)$$

where F is a free energy offset.

To combine the biased distributions from all windows and reconstruct the full PMF, the Weighted Histogram Analysis Method (WHAM) is employed. WHAM solves self-consistent equations to combine histograms while accounting for the biasing potential applied in each window. The final result is a globally consistent estimate of $G^u(\xi)$ over the entire range of sampled coordinates.

3 Methods

3.1 System Setup

The protein used in this study was the KcsA potassium channel, embedded in a pre-equilibrated DPPC lipid bilayer using the

CHARMM-GUI Membrane Builder. The membrane patch consisted of 112 lipid molecules with the protein inserted and aligned along the bilayer normal (z -axis). The system was solvated with 10203 TIP3P water molecules and neutralized by adding 30 potassium and 42 chloride ions to maintain a physiological salt concentration of 0.15 M. Initially, 3 potassium ions are placed in selectivity filter (including one in cavity).

All topology and coordinate files were generated using the CHARMM36m force field for proteins and lipids.

3.2 Energy Minimization and Equilibration

Energy minimization was first performed using the steepest descent algorithm until the maximum force fell below 1000 kJ/mol/nm. This step ensured removal of any steric clashes in the system.

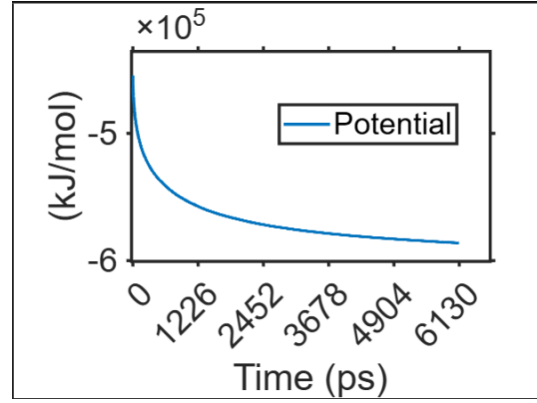


Figure 3. Potential energy of the system as a function of simulation time during the equilibration phase. The steady decrease in potential energy over time indicates system stabilization. Final value = -568381 kJ/mol

Following minimization, a position-restrained annealing step was performed, where the system was gradually heated from 0 K to 310 K over a period of over 1 ns. This process helped relieve steric clashes and gently brought the system to the target temperature while avoiding structural distortions.

Subsequently, a six-step equilibration protocol from CHARMM-GUI was followed. Each step included position restraints on different components (protein, lipids, and ions), with gradually decreasing force constants. The equilibration stages involved:

- 125 ps NVT equilibration at 310 K using the V-rescale thermostat.
- Five consecutive NPT equilibration steps totaling 1.375 ns using the Parrinello-Rahman barostat to maintain 1 bar pressure.

All bonds involving hydrogen atoms were constrained using the LINCS algorithm. A cutoff of 1.2 nm was applied for van der Waals and Coulomb interactions, and long-range electrostatics were treated with the Particle Mesh Ewald (PME) method.

Following equilibration, production runs and umbrella sampling were initiated using restrained center-of-mass pulling simulations, as detailed in the next section.

3.3 Umbrella Sampling Setup

To investigate the free energy profile of potassium ion conduction through the KcsA channel, umbrella sampling was used along the reaction coordinate defined as the z -axis distance between the potassium ion and the center of mass (COM) of the protein. The initial configurations for the umbrella windows were generated from a steered molecular dynamics (SMD) pulling simulation, in which a single potassium ion was pulled through the filter at a constant force in each window along the z -axis using a harmonic biasing potential:

$$V(z) = \frac{1}{2}k(z - z_0)^2$$

where k is the force constant (typically 300–1000 kJ/mol/nm²) and z_0 is the center of the window.

Snapshots at regular intervals (e.g., 0.1 nm spacing) along the pulling path were extracted and used to initiate umbrella sampling windows. Each window was simulated independently with the harmonic potential restraining the ion at the specified z_0 position.

For each window, simulations of 100–200 ps were performed in the NPT ensemble. The biasing force constant was tuned (between 300–1000 kJ/mol/nm) after each window to make sure that spacing did not exceed more than 0.1 nm, which is necessary to get better overlap between neighboring histograms.

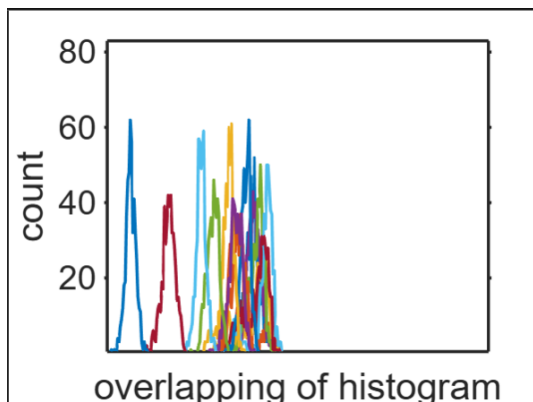


Figure 4. Overlapping of sampling is being done at important part of filter.

The potential of mean force (PMF) was reconstructed from the umbrella histograms using the Weighted Histogram Analysis Method (WHAM), as implemented in GROMACS.

4 Results

Following system equilibration, the potassium ions spontaneously adopted a configuration within the selectivity filter where they preferentially occupied the S1, S3, and S4 binding sites. This ion distribution remained stable throughout the NPT equilibration phase and served as the starting point for subsequent umbrella sampling simulations.

This configuration aligns with the known alternating ion-water occupancy model and is consistent with prior structural and simulation studies of the KcsA channel. The stability of the S1–S3–S4 occupancy suggests that the energy landscape favors this arrangement under the simulated conditions.

To explore the free energy profile of ion conduction, umbrella sampling was performed using the S4–S3–S1 ion configuration as

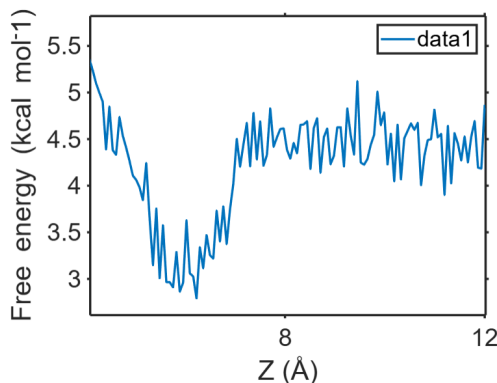


Figure 5. Energy minima before 8 Å(S0 site). Extracellular side as z increases

the initial state. The position of a single ion was perturbed along the z -axis while the other two ions were restrained at their equilibrated locations. The PMF profile was reconstructed to capture the energy barriers and favorable binding sites during ion conduction through the selectivity filter.

Potential of Mean Force Analysis

The data have little disturbance. Disturbance could be reduced by increasing more number of umbrella sampling windows and simulation time of each window. Due to time constraint, simulation is done for just one state.

The one-dimensional PMF profile along the Z -axis, representing the coordinate of the permeating potassium ion through the KcsA selectivity filter, reveals several energy minima and barriers indicative of preferred ion-binding sites and transition states, respectively (see Figure 5). The free energy decreases from approximately 5.4 kcal/mol at $Z = 4.1$ Å to a minimum of around 2.86 kcal/mol near $Z \approx 5.8$ Å, corresponding to a deep energy well. This likely represents a stable ion-binding site within the selectivity filter, possibly at position S0.

Overall, the PMF profile supports a stepwise conduction mechanism through the selectivity filter, with energetically favorable binding sites spaced along the reaction coordinate. The magnitude of the barriers and wells observed is consistent with known knock-on or soft knock-on mechanisms(mentioned in paper), which require low-to-moderate energy barriers for physiological ion conduction through the KcsA channel.

5 Discussion

The one-dimensional potential of mean force (PMF) profile reveals a relatively smooth free energy landscape along the chosen reaction coordinate for ion movement through the K⁺ channel. The minima observed between 5.6 to 6.3 Å correspond to the most thermodynamically favorable binding regions, likely indicating stable occupancy sites such as S4 or adjacent vestibular zones.

The free energy barrier near 5.4 Å followed by a sudden drop to 5.6 Å suggests the transition state region for ion entry into the selectivity filter. The fluctuations observed beyond 6.3 Å represent relatively shallow variations with no deep well, indicating fewer stable interactions or less frequent sampling in those regions.

Moreover, our umbrella simulations were initiated from a con-

figuration in which ions occupied S1, S3, and S4 sites after equilibration. This biased occupancy may have influenced the free energy landscape by stabilizing certain intermediate configurations over others. Consequently, the PMF should be interpreted as conditional to this initial state, and additional simulations from alternative occupancies could help validate or extend these findings.

Overall, the computed PMF shows qualitative agreement with trends observed in previous studies of the KcsA channel [Simon Bernèche, 2001](#), supporting the two-step knock-on mechanism with alternating stable and transition zones. Further work can involve extending the sampling toward positioning of ions at different position initially.

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